Atoms and elements

Elements

An **element** is a simple substance that cannot be split into anything simpler by chemical reactions. Atoms are the smallest particles of an element that can exist. **Atoms** of one element are all the same, and are different from atoms of all the other elements.
There are over 100 different elements. All the elements are shown in the **Periodic Table**. Each element has a **chemical symbol**, which is usually one or two letters. A symbol is written with the first letter as a capital, and the second letter is small.

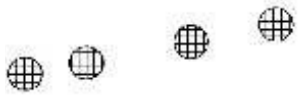
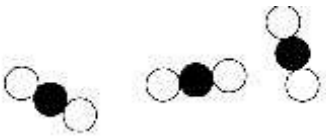
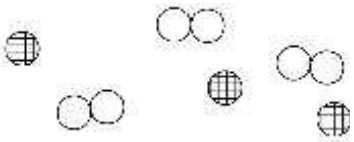
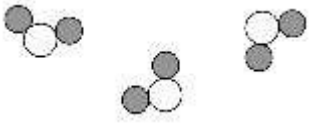
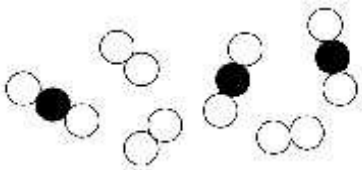
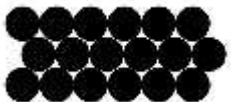
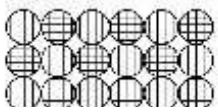
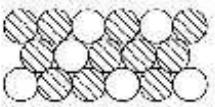
carbon	C	oxygen	O
nitrogen	N	hydrogen	H
gold	Au	silver	Ag
copper	Cu	aluminium	Al

Compounds

Elements can join together to make compounds. The name of the compound tells you the elements that are in it. Compounds made from two elements always have a name which ends in '**-ide**'.

These elements join together …	**… to make these compounds**
carbon, oxygen	carbon dioxide
sodium, chlorine	sodium chloride
magnesium, oxygen	magnesium oxide

A chemical formula tells you the name and number of atoms in a compound. The smallest particle of many compounds is called a **molecule**. Molecules are made up of atoms. Some elements are also made of molecules. For example, a molecule of oxygen contains two oxygen atoms joined together. The formula is O_2.

Elements	**Compounds**	**Mixtures**
Atoms of helium (He) 	Molecules of carbon dioxide (CO_2) 	A mixture of helium and oxygen 
Molecules of oxygen (O_2) 	Molecules of water (H_2O) 	A mixture of carbon dioxide and oxygen 
A lump of carbon 	A lump of sodium chloride (NaCl) 	A lump of bronze(an alloy of copper and tin) 

ATOMS AND ELEMENTS QUIZ

1. Which of these things is an element?
A salt
B water
C oxygen
D air

2. The number of chemical elements in the world is:
A 4.
B about 100.
C about 1000.
D over a million.

3. The element that makes up most of the mass of your body is:
A oxygen.
B carbon.
C calcium.
D hydrogen.

4. Chemical elements:
A are always solids.
B are either solids or liquids.
C are always gases.
D can be solids, liquids or gases.

5. Which of these metals is not a pure element?
A gold
B copper
C silver
D bronze

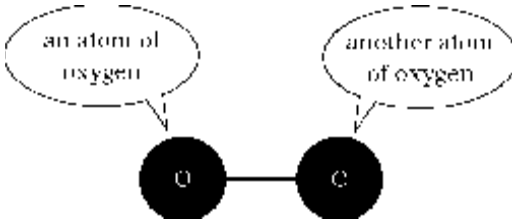
6. The smallest part of an element is called:
A an atom.
B a molecule.
C a compound.
D an ion.

7. An element is something that:
A can be split up by a chemical reaction.
B cannot be split up by a chemical reaction.
C is made up of different kinds of atom.
D is a mixture of atoms and molecules.

8. The correct way to write the chemical symbol for calcium is
A CA
B Ca
C ca
D Ca

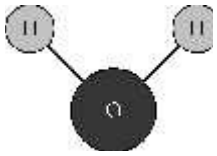
Compounds and mixtures

Elements are simple substances which cannot be split up in chemical reactions. **Atoms** are the smallest particles of an element that can exist. Atoms of an element are all the same.
Each element has its own chemical symbol. For example, the **chemical symbol** for oxygen is O.
Some elements have their atoms joined to each other in small groups called **molecules**. Oxygen is an example.

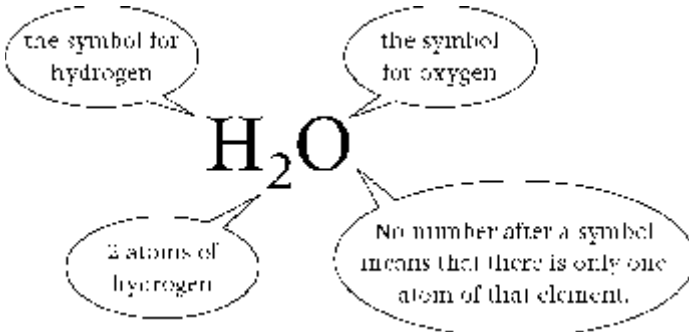


Compounds

Elements can join together to make **compounds**. A compound contains two or more elements joined together. The name of the compound tells you the elements that are in it. Compounds made from two elements always have a name which ends in '-ide'.
Many compounds exist as atoms attached to each other in small groups – molecules.



The chemical formula tells you the number of atoms of each element in a compound. Each element in the formula is shown by its chemical symbol. For example:



A compound always contains the same elements in the same ratio.

The properties of a compound are different from the elements that make it up. For example, hydrogen is an explosive gas and oxygen will relight a glowing splint but water is a liquid which will put fires out.

Chemical reactions

Compounds can react chemically by mixing them with other chemicals, or by using heat or electricity. You can tell that a chemical reaction has occurred if there is a colour change or when a gas is given off.

Most chemical reactions also involve an energy change. This is usually in the form of heat, but can also involve light being given off, for example, in burning (combustion).

In a chemical reaction a new substance is always formed. Most chemical reactions are not easily reversed (they are irreversible).

Some chemical reactions take place just by mixing. When you make a solid by mixing two liquids, the solid is called a precipitate.

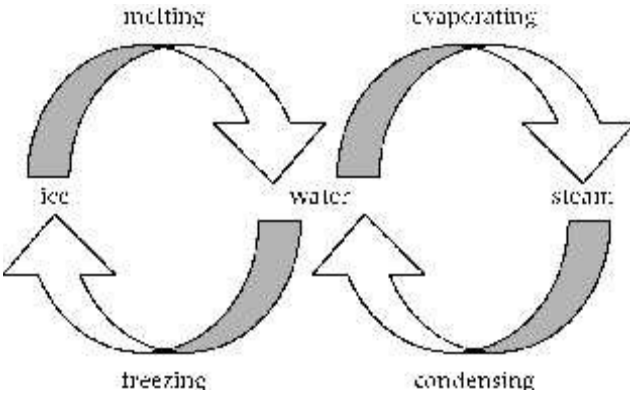
We can write **word equations** to show a chemical reaction. The chemicals that you start with are called the **reactants**. The chemicals at the end are called the **products**. For example:

Magnesium + Oxygen $\longrightarrow$ Magnesium oxide

Reactants **Products**

Physical changes

In a **physical change** no new substance is formed. **Melting**, **evaporating**, **condensing** and **freezing** are all examples of physical changes. For example:



Mixtures

Elements and compounds can also be mixed together. A **mixture** is easier to separate than the elements in a compound. Soil, river water and sea water are examples of mixtures that occur naturally.

Elements and compounds melt and boil at a fixed temperature. Mixtures do not have definite **melting points** and **boiling points**.

Air is a mixture of gases most of which is nitrogen and oxygen. The gases in the air can be separated by fractional distillation.

Compounds and mixtures quiz

1 An atom is:
A the smallest particle that can exist.
B a group of particles that are all the same.
C the smallest particle of a compound that exists under normal conditions.
D the smallest particle of an element that exists under normal conditions.

2 A molecule is:
A the smallest particle that can exist.
B the smallest particle of an element that can exist.
C two or more atoms chemically joined together.
D the same as an atom.

3 The number of types of atom in a compound is:
A only one.
B one or two.
C two or more.
D three or more.

4 How many different elements are there in the compound with the formula NH3?
A one
B two
C three
D four

5 Which of these things does **not** show that a reaction has occurred?
A The mixture gets hotter.
B The substance changes from a liquid to a solid.
C The mixture changes colour.
D Bubbles of gas form in the mixture.

6 The substances that you get at the end of a chemical reaction are called:
A reactants.
B chemicals.
C products.
D gases.

7 Which of these changes is a chemical reaction?
A Melting some margarine in a pan.
B Boiling a kettle of water.
C Frying an egg.
D Cooling a drink by putting an ice cube in it.

8 Complete this word equation:
sodium carbonate + copper sulphate ® sodium sulphate + _____
A copper sulphate
B copper oxide
C sodium carbonate
D copper carbonate